

# Cheedella Vamsi Kishore

Hyderabad, India

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## PROFESSIONAL SUMMARY

AI Engineer with **3.5 YOE**, specializing in **agentic LLM systems, browser & web automation agents, production RAG, and LLMOps**. Currently sole AI Engineer at Deep Algorithms driving **0→1 build** of a browser-controlling autonomous pentesting agent (LangGraph + ReAct + MCP + multi-LLM routing; **~80% detection** across 10 attack classes), now in customer pilot. Previously led AI engineering on a 3-person team at Capgemini, shipping a 6-feature GenAI Scrum-Master assistant across 3 enterprise clients (**~45% reduction** in manual overhead). Strong focus on LLMops, evaluation (DeepEval, Ragas, LangFuse), and cost-aware inference routing.

## EXPERIENCE

### AI Engineer

Deep Algorithms Solutions Pvt Ltd

Sept 2025 - Present

Hyderabad, India

**Autonomous AI Agent for Web Application Testing. (DAST)** Browser-controlling agent built on Playwright + LangGraph + ReAct + MCP, executes multi-step reasoning workflows over live web applications (security testing as primary use case).

- Led **0→1 build** of an **autonomous browser-controlling AI agent** (Playwright + LangGraph + ReAct, MCP-isolated tools) as sole AI Engineer, market research to pilot in 5 months — agent navigates live web applications, parses DOMs, and executes multi-step reasoning workflows over them. Owned architecture end-to-end: multi-LLM orchestration, MCP tool isolation, 3-layer memory, evaluation harness.
- Validated across **60+ scan runs** on Altoero Mutual and Acunetix benchmarks — **~80% detection** across **10 attack classes on 5 surfaces**; currently in customer pilot under NDA.
- Designed **dual-LLM routing** — Claude Opus (ReAct + CoT) for stateful reasoning attacks, **Llama 3.1 70B via AWS Bedrock** for WAF-adaptive payload mutation; benchmarked multiple Bedrock-hosted models before settling on Llama 3.1 — cut per-scan LLM cost **~30%** vs all-frontier-API baseline while preserving detection rate.
- After prototyping with LangGraph, CrewAI, and a custom loop, Shipped a **LangGraph ReAct** agent over an isolated MCP topology — each attack surface acts as a scoped client with dedicated MCP servers; servers are visible only to their owning client, preventing tool bleed across surfaces.
- Architected **pentester-extensible system** — git-versioned **skills** (YAML + Playwright), lifecycle **hooks** for session/CSRF/env updates, and plugin-style **MCP composition**; pentesters ship new attack classes by authoring skills, not Python. **12 skills shipped** (SQLi, XSS, IDOR, broken auth, SSRF + recon) by pilot.
- Architected 3-layer memory — **Episodic** (Neo4j temporal graph, 50 reasoning steps/scan; Cypher correlates findings into attack chains); **Semantic** (Qdrant hybrid BM25 + dense, surfaces WAF mutation library); **Procedural** (pentester-authored skills). Paired with dual-state environment — mutable auth tokens, immutable recon metadata.
- Episodic memory surfaced **multi-step attack chains** (e.g., SQLi + IDOR → account takeover) in ~4 of 12 sampled pilot runs — findings stateless agents miss by construction, now visualized as Neo4j attack-chain graphs in the PDF reports.
- Built **recon pipeline** feeding immutable environment state: Nmap + Subfinder (ports/subdomains), OWASP ZAP scans mapped to attack surfaces, and custom Playwright crawlers with DOM parsing + directory bruteforcing — covering authenticated and unauthenticated routes; auth layer preserves CSRF, session cookies across scan lifecycle.
- Cut false-positive rate from **~40% to ~24%** by adding **LLM-as-Judge validation** on top of Playwright verification — evidence-grounded cross-checking of HTTP response bodies for reasoning attacks.
- Broader eval framework tracks **Detection Rate, Tool Selection, Memory Recall, and Reasoning Chain Coherence** (security-team scored); outputs structured PDF reports with repro steps and Neo4j attack-chain visualizations.
- Deployed on **AWS** (Dockerized FastAPI via ECR to EC2, artifacts to S3, Bedrock for Llama inference); hybrid inference routing (Bedrock Llama + Anthropic API Claude + stateless EC2) holds per-scan cloud spend at **~\$15**. Average end-to-end scan: **~40 min**.

### Data Scientist - Associate Consultant

Capgemini

Sept 2022 – Sept 2025

Hyderabad, India

**Multi-Feature GenAI Product for Enterprise Agile Workflows.** Production RAG, Document Intelligence, and LLM-driven Automation over Azure DevOps Artifacts.

- Cut Scrum-Master manual overhead **~45% on large teams and ~30% on small teams** (measured via post-deployment feedback from ~10 SMs across **3 clients**) by shipping a **6-feature GenAI assistant** over Azure DevOps artifacts; owned architecture, retrieval design, prompt strategy, and evaluation as **senior IC on a 3-person team**.

- Shipped **InsightFlow** document intelligence module with **sliding-window chunking (20% overlap)**; backed by ChromaDB or Azure AI Search depending on client preference; evaluated output quality via DeepEval.
- Built **Ask Doc**, a conversational RAG layer over user stories, test cases, and defects (GPT-4o + LangChain) — moved from naive to **hybrid retrieval** after A/B evaluation on a ~50-question eval set showed substantial gains in contextual recall and faithfulness.
- Scaled test-case scoring  $\sim 5\times$  over the sequential PoC by building **async parallel GPT-4o ranking** across Complexity, Dependency, and Criticality axes — ran over ~300 test cases across 3 clients.
- Deployed **duplicate defect/test-case detection** using Azure OpenAI embeddings with cosine similarity thresholds tuned against QA-labeled ground truth, across **300+ test cases**; eliminated manual dedupe triage.
- Built **evaluation harness using DeepEval** spanning 3 feature modules — Faithfulness + Answer Relevancy for document QA, Contextual Precision/Recall + Hit Rate for retrieval, Precision/Recall/F1 against QA-labeled ground truth for deduplication; evaluation signals drove the naive → hybrid retrieval migration and threshold tuning for similarity matching.
- Engineered **role-aware RCA** (Developer / QA / PM prompt personas) over historical defects and a **sprint-artifact generator** configurable to each client's Agile template — enabled **per-user config** (RAG params, prompts, feature toggles) so teams tailored behavior without code changes.

## TECHNICAL SKILLS

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| <b>Languages</b>                    | : Python, SQL                                                                                                                             |
| <b>AI/ML Frameworks</b>             | : PyTorch, TensorFlow, HuggingFace Transformers, Scikit-Learn, pandas, NumPy                                                              |
| <b>LLM &amp; Agentic Frameworks</b> | : LangChain, LangGraph, CrewAI, MCP (stdio / HTTP / SSE), ReAct, Tool-Calling, vLLM, Ollama, KV Caching                                   |
| <b>Browser &amp; Web Automation</b> | : Browser Agents, Autonomous Web Agents, Web Application Automation, DOM Parsing                                                          |
| <b>LLMs &amp; Fine-Tuning</b>       | : Claude, GPT-4 / 4o, Llama (via Bedrock), Mistral, Qwen, RoBERTa, BERT   PEFT, LoRA, QLoRA                                               |
| <b>RAG &amp; Retrieval</b>          | : Hybrid Retrieval (BM25 + Dense), Reranking (Cross-Encoder), Chunking Strategies, Agentic RAG, GraphRAG, Knowledge Graph Memory          |
| <b>Vector &amp; Graph DBs</b>       | : Qdrant, ChromaDB, Neo4j (Graph DB)   Azure SQL                                                                                          |
| <b>LLMOps &amp; MLOps</b>           | : DeepEval, Ragas, MLflow, LangFuse, LangSmith   Metrics: Faithfulness, Answer Relevancy, Contextual Precision/Recall, Hallucination Rate |
| <b>Backend &amp; APIs</b>           | : FastAPI, Flask, Streamlit, RESTful APIs, Pydantic, Playwright                                                                           |
| <b>Cloud &amp; DevOps</b>           | : AWS (EC2, S3, ECR, Bedrock), Azure (OpenAI, Blob Storage, App Service), Docker, Git, Linux                                              |

## PROJECTS

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### RAG Benchmark Harness | *Llama 3.1, Claude, LangFuse, LangSmith, MLflow, DeepEval, Ragas, Docker* [code]

- Built **end-to-end RAG benchmark harness** on India Union Budget 2024-25 — modular pipeline (chunking, dense + BM25 + hybrid retrieval, cross-encoder reranking, versioned YAML prompts) benchmarking **10 configurations** across chunk size, retrieval, reranking, and models (Llama 3.1 / 3.3 / Claude Haiku) on a 25-question eval set.
- Applied **LLMOps** practices throughout — **LangFuse + LangSmith** tracing per run, **MLflow** for experiment tracking, **DeepEval + Ragas** scoring on Faithfulness, Answer Relevancy, Contextual Precision/Recall.

### LLM Fine-Tuning: PEFT Efficiency Study | *PyTorch, HuggingFace, PEFT, LoRA, QLoRA, LLaMA-2, RoBERTa*

- Hands-on exploration of parameter-efficient fine-tuning — **QLoRA on LLaMA-2-7B** (adapted ~0.29% of parameters, 19M / 6.7B) for causal language modeling, and **LoRA on RoBERTa** (adapted ~0.52% of parameters, 665K / 125M) for SST-2 sentiment classification; demonstrated memory-efficient domain adaptation without full-model compute cost.

## EDUCATION

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### National Institute of Technology, Calicut

B.Tech Electrical and Electronics Engineering, CGPA: 8.17/10

Calicut, Kerala

Aug 2018 – May 2022

## CERTIFICATIONS

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- **Anthropic:** Introduction to Agent Skills, AI Fluency for Students
- **Coursera:** Deep Learning Specialization (Andrew Ng, deeplearning.ai)